Multiphase Flows – Laboratory Session 4:
Spray A: Euler-Lagrange CFD Simulation of a High-Pressure Diesel Spray

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Methodology

The Engine Combustion Network (ECN) *Spray A* is a high-pressure ($p_{\text{inj}} = 150$ MPa), single-hole, axially oriented diesel nozzle ($d_0 = 0.09$ mm) with *n*-dodecane ($C_{12}H_{26}$) as a diesel surrogate [1]. The ambient conditions assigned to this group are $p_{\text{amb}} = 7$ MPa and $T_{\text{amb}} = 733.33$ K, derived from the formula in the statement. These conditions differ from the ECN nominal reference ($T_{\text{amb}} = 900$ K, $p_{\text{amb}} = 6$ MPa), so differences in liquid length and vapour penetration are expected.

A Discrete Droplet Model (DDM) Euler-Lagrange framework is employed in Simcenter StarCCM+. The carrier gas (air + dodecane vapour, treated as a multi-component ideal gas) is solved as a continuum Eulerian phase via the Reynolds-Averaged Navier-Stokes (RANS) equations with the standard k-ε turbulence model. Liquid droplets are tracked as Lagrangian parcels, each representing a statistical ensemble of identical droplets. A four-way coupling strategy is used: momentum, energy and mass are exchanged between phases (two-way coupling), and inter-parcel collisions are handled via NTC model. The governing transport equations for species k and momentum in the Eulerian phase include Lagrangian source terms $\dot{m}_k$ and $\mathbf{S}_{\text{mom}}$ arising from evaporation and drag forces, respectively:

$$\frac{\partial(\rho Y_k)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_k) = \nabla \cdot (\rho D_k \nabla Y_k) + \dot{m}_k, \qquad \frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{S}_{\text{mom}}. \tag{1}$$

Primary atomisation is handled by the Huh atomisation model, which accounts for turbulence-induced breakup at the nozzle exit. Secondary breakup follows the KH-RT model: Kelvin-Helmholtz instabilities govern the near-nozzle region, while Rayleigh-Taylor instabilities dominate once sufficient droplet deceleration is reached. Evaporation is modelled via Quasi-Steady Evaporation, driven by the vapour concentration gradient between the droplet surface (saturation condition) and the far-field gas. Drag is computed with the standard spherical-drag law, and turbulent dispersion accounts for the stochastic interaction of parcels with turbulent eddies of the carrier phase.

The domain is a cylinder of radius 25 mm and length 100 mm. An unstructured polyhedral mesh (base size 2 mm, growth rate 1.05) is used with two conical refinement zones along the spray axis: cell sizes of 0.25 mm and 0.5 mm in the inner and outer cones, respectively, yielding ≈324,000 cells. The PISO unsteady solver is run with an adaptive time-step (target mean CFL 0.5, max. CFL 2.0; $\Delta t_{\text{min}} = 10^{-7}$ s, $\Delta t_{\text{max}} = 10^{-4}$ s) for 2 ms of physical time. Liquid and vapour penetrations are monitored online using the ECN 99% criterion (axial distance containing 99% of the liquid/vapour mass).

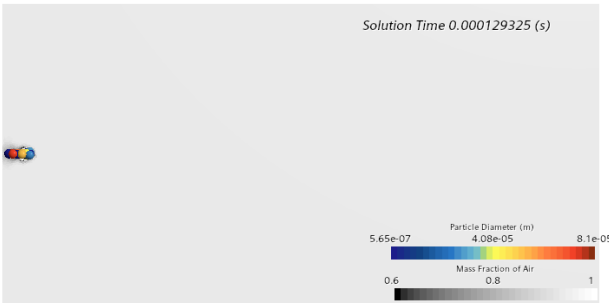
Results and Conclusions

Figure 1 shows the mid-plane air mass fraction (grey scale) together with the Lagrangian parcels coloured by particle diameter at four representative instants. A full temporal animation of the injection event is available.

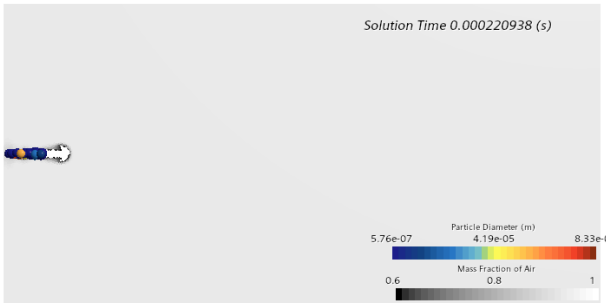
At $t \approx 0.13$ ms (Figure 1a), the droplet population near the nozzle exit already spans a wide diameter range: large parent parcels (orange/red, $d40\,\mu$m) coexist with sub-micron KH-RT child droplets (blue), but no vapour plume is visible yet. By $t \approx 0.22$ ms (Figure 1b), a visible vapour halo surrounds the droplet cluster, confirming that evaporation is already active at this very early stage (a direct consequence of the high ambient enthalpy at 733 K). As injection proceeds, the liquid core stabilises at the quasi-steady liquid length while the vapour plume continues to grow axially, clearly visible as the darker grey region in Figure 1(c) at $t \approx 1.0$ ms. After injection ends (Figure 1d), no significant liquid phase remains and the vapour cloud disperses driven by diffusion and residual momentum.

Figure 2 shows the mid-plane axial velocity field at three instants. At $t \approx 0.29$ ms the jet is compact with peak velocities up to 207 m/s and a thin negative-velocity fringe at the spray edges already marking the onset of air entrainment. By $t \approx 1.5$ ms the momentum-carrying region has broadened considerably, consistent with the radial profiles in Figure 3b. After injection ends ($t \approx 2.0$ ms, shown with iso-contour lines), the colour scale compresses to just 32.2 m/s, illustrating the rapid momentum dissipation; two small symmetric recirculation lobes (≈ -2 m/s) flank the residual vapour cloud as entrained air rebounds once the driving parcels are gone.

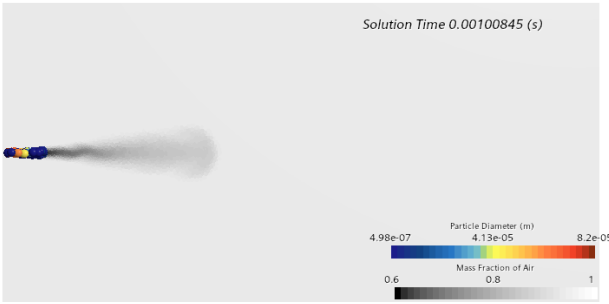
The high ambient temperature ($T_{\text{amb}} = 733$ K) drives intense evaporation from the earliest instants, so even at $t = 0.13$ ms the droplet population already spans a wide diameter range: large parent parcels near the nozzle and tiny child droplets produced by KH-RT breakup and partial evaporation further downstream. By $t = 0.22$ ms (Figure 1b), a visible vapour halo surrounds the droplet cluster, confirming that evaporation is already active at this very early stage (a



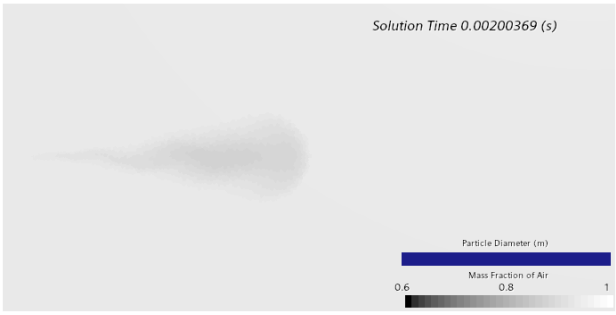
(a) $t \approx 0.13$ ms. Early injection; no vapour plume visible yet.



(b) $t \approx 0.22$ ms. Onset of evaporation; first vapour halo appears.

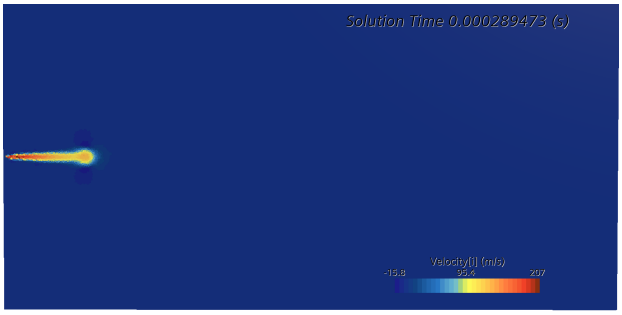


(c) $t \approx 1.0$ ms. Fully developed spray; stable liquid length.

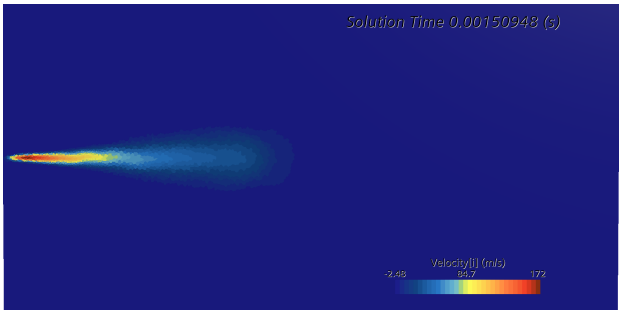


(d) $t \approx 2.0$ ms. Post-injection; vapour cloud disperses radially.

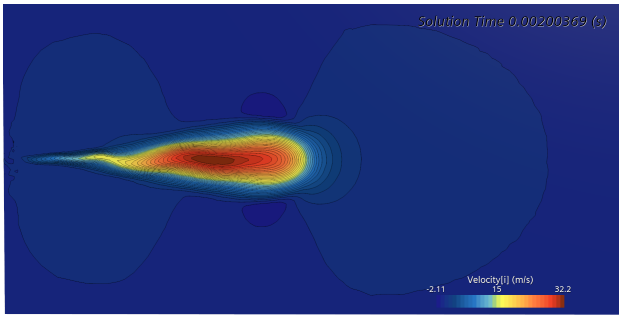
Figure 1: Mid-plane air mass fraction (grey-scale) overlaid with Lagrangian parcels coloured by particle diameter at four simulation instants ($p_{\text{amb}} = 7$ MPa, $T_{\text{amb}} = 733$ K). Full temporal evolution available in the video.



(a) $t \approx 0.29$ ms. High-speed core up to 207 m/s; entrainment fringe (≈ -16 m/s) at spray edges.



(b) $t \approx 1.5$ ms. Wider, more diffuse momentum plume (up to 172 m/s); radial mixing evident downstream.



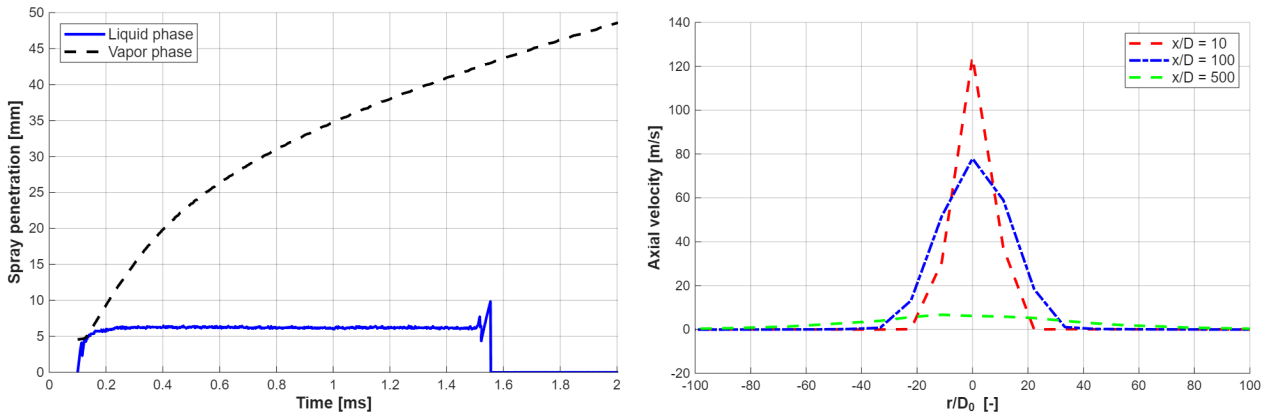
(c) $t \approx 2.0$ ms (iso-contours). Scale compressed to 32.2 m/s; recirculation lobes (≈ -2 m/s) flank the vapour cloud.

Figure 2: Mid-plane axial velocity field ($p_{\text{amb}} = 7$ MPa, $T_{\text{amb}} = 733$ K) at three instants spanning the injection event.

direct consequence of the high ambient enthalpy). As the injection proceeds, the liquid core stabilises at the quasi-steady liquid length while the vapour plume continues to grow axially, clearly visible as the darker grey region in Figure 1(c). After injection ends (Figure 1d), no significant liquid phase remains and the vapour cloud disperses driven by diffusion and residual momentum.

Figure 3a shows both penetrations following the ECN 99% criterion. The liquid penetration rises steeply during the first $\sim$0.2 ms and then stabilises at a quasi-steady liquid length of LL $\approx$ 9 mm, after which evaporation exactly balances the incoming liquid mass flux. This is noticeably shorter than typical ECN results at lower ambient densities,

which is consistent with the increased aerodynamic drag imposed by the denser ambient at 7 MPa: droplets decelerate faster and have less time to penetrate before being fully evaporated. The moderate ambient temperature (733 K vs. the 900 K reference) partially counteracts this by slowing evaporation, but the pressure effect dominates. A small numerical artefact is visible around $t \approx 1.6$ ms in the liquid curve (likely caused by a sudden change in parcel count near the end of injection) but it does not affect the physical plateau value. The vapour phase, by contrast, penetrates continuously throughout the entire injection event, reaching approximately 47 mm at $t = 2$ ms. Its growth rate slows progressively as the driving momentum decreases after injection ends, broadly consistent with the expected $L_v \propto t^{1/2}$ scaling in the momentum-driven regime. It is worth noting that the vapour penetration curve shows no plateau, confirming that the vapour front is still advecting and diffusing at the end of the simulation window.



(a) Liquid (solid) and vapour (dashed) axial penetration vs. time. The liquid length plateau (≈ 9 mm) is established by $t \approx 0.2$ ms and maintained throughout injection.

(b) Radial profiles of axial velocity at $x/D_0 = 10$, 100, 500 at $t = 1.5$ ms. Peak velocity decays strongly with distance as momentum transfers to the entrained air.

Figure 3: Spray A macroscopic characteristics ($p_{\text{amb}} = 7$ MPa, $T_{\text{amb}} = 733$ K).

Figure 3b presents the axial velocity radial profiles at $x/D_0 = 10$, 100, and 500 at $t = 1.5$ ms. The profile at $x/D_0 = 10$ is sharp and jet-like, with a peak of roughly 120 m/s, what is significantly below the nominal injection velocity of 633 m/s, indicating that substantial momentum has already been transferred to the gas within the first millimetres downstream of the nozzle. This rapid decay is a consequence of the high ambient density (7 MPa), which increases the aerodynamic resistance experienced by the parcels. At $x/D_0 = 100$ the peak drops to approximately 65 m/s and the profile broadens considerably, reflecting ongoing turbulent mixing and air entrainment. By $x/D_0 = 500$ the profile is nearly flat at around 8 m/s, consistent with the spray having largely diluted into the ambient. The profiles are symmetric about the axis, as expected for this axisymmetric configuration, and all decay smoothly without secondary peaks, suggesting that the mesh resolution in the outer refinement cone is sufficient to capture the mean shear layer. One limitation of this plot is that the $x/D_0 = 1$ location is absent (likely because it falls within the injector region where parcel initialisation dominates over the fully developed flow field) so the very near-nozzle behaviour cannot be assessed directly from this figure.

Conclusions. An Euler-Lagrange DDM simulation of the ECN Spray A nozzle was successfully set up and run under $p_{\text{amb}} = 7$ MPa and $T_{\text{amb}} = 733.33$ K. The liquid penetration rapidly reaches a quasi-steady liquid length of ≈ 9 mm and remains there throughout the injection event, controlled primarily by the high aerodynamic drag at 7 MPa rather than by evaporation rate. The vapour phase grows continuously to approximately 47 mm at 2 ms, with no sign of levelling off at the end of the simulation. The velocity field confirms strong air entrainment at the spray periphery (evidenced by small negative axial velocities) which is the dominant mechanism driving vapour dilution and mixing. The radial profiles show that most of the injected momentum is transferred to the ambient within the first $x/D_0 \approx 100$, highlighting how aggressively the high-density environment decelerates the spray. Together, these results are physically coherent and qualitatively consistent with the expected behaviour of a high-pressure diesel spray under engine-like ambient conditions.

References

[1] Engine Combustion Network, *Spray A specifications*, `https://ecn.sandia.gov`, 2023.
[2] Baumgarten, C., *Mixture Formation in Internal Combustion Engines*, Springer, 2006.